EV-Auth: Lightweight Authentication Protocol Suite for Dynamic Charging System of Electric Vehicles With Seamless Handover

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***Abstract*—Electric vehicle adoption has accelerated globally, necessitating the development of an efficient charging technology. Static and partially dynamic charging are currently in use, but such types of charging reduce vehicles' ability to stand charge, contributing to range anxiety. Dynamic charging of electric vehicles will be a promising technology in the near future, allowing an electric vehicle to charge by itself while in motion. However, assessing the security risks before enabling dynamic charging is mandatory, with authentication being a necessary step. Hence, a protocol suite EV-Auth is proposed for communicating entities in a dynamic charging system capable of mutual authentication and session key computation. To achieve practical capabilities, EV-Auth is built using lightweight cryptographic primitives. Additionally, it includes a unique feature known as seamless electric vehicle handover. The security of the proposed protocol suite is assessed both informally and formally using random oracle model and scyther tool. The performance analysis demonstrates that the proposed protocol suite satisfies the communication and computation time requirements.**

***Index Terms*—Electric vehicles, dynamic charging, authentication protocol, handover, random oracle model, scyther tool, security.**

I. Introduction

ELECTRIC vehicles (EVs) have made significant advancements in recent years in terms of performance and range, and are rapidly gaining automotive technology leadership. EVs have the potential to reduce reliance on oil, thereby mitigating the long-term impact of global CO2 emissions on global warming [1]–[3]. The adoption of EVs has spawned a widespread human phenomenon known as range anxiety, which would be exacerbated by a lack of commercial charging infrastructure within EVs' current driving range [4].

As the popularity of EVs as an alternative mode of transportation has grown, the ability to charge the vehicle away from home has become a critical issue. Wireless charging enables three modes of EV charging: static charging, quasi-dynamic charging, and dynamic charging. Static charging was intended to charge EVs in public parking lots while the vehicles were not in use for an extended period of time. The quasi-dynamic wireless charging system recharges EVs while they are stationary for a brief period of time, such as at traffic lights. Dynamic wireless charging refers to the process of powering an electric vehicle while it is in motion. The EV's battery is recharged while it is in motion via wireless charging pads embedded in the road surface [5]–[7]. Dynamic charging is a promising technology that has the potential to alleviate the world's long charging times by allowing EVs to charge while on the road. This is accomplished by embedding charging devices known as charging pads (CPs) beneath the road that recharge EVs via electromagnetic energy stimulation. Hence, dynamic charging technology eliminates the need for EVs to stop and recharge their batteries at charging stations, saving EV drivers time significantly [8], [9].

Nevertheless, there are still unaddressed constraints to dynamic charging systems, as their deployment requires extensive collaboration among several communicating principles, including EVs, highway infrastructure, and charging stations. Additionally, a dynamic charging system has security and communication issues, such as message tampering, spoofing, or delays among other things, to disrupt the charging service. The EV dynamic charging system's efficacy is determined by the frequency with which various robust security mechanisms such as authentication, data integrity, message confidentiality, forward secrecy, and user anonymity are achieved [10]. Otherwise, an adversary can intercept/interrupt the dynamic charging system through a variety of attacks, including impersonation, password guessing, denial-of-service (DoS), replay, and man-in-the-middle [11].

Considering the complexity of communications between EVs and other involved entities, security researchers have proposed numerous authentication protocols [12]–[18] for the EV dynamic charging system. The authentication protocols enable entities to authenticate their connection to the charging station

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(Fog Server (FS), Roadside Unit (RSU), or CP) and to verify the charging station's authenticity. Each entity within the charging station verifies the EV charging request and dynamically accepts the EV to be induced through the charging lane.

While several types of protocols exist for the electric vehicle dynamic charging system, they are either insufficiently secure or underperforming. Additionally, certain protocols are decentralised in nature and built on top of the most powerful cryptographic primitives such as blockchain that raises concerns about implementations. Motivated from the aforementioned works, a robust and lightweight mutually authenticated key agreement protocol suite with smooth handover is proposed for an EV dynamic charging system. The main contributions can be summarized as follows:

- In this paper, critical security challenges and requirements for EV dynamic charging systems are analyzed and also, unaddressed key constraints are identified.
- A robust and lightweight mutually authenticated key agreement protocol suite for EV dynamic charging, EV-Auth, is proposed that incorporates several novel features, including seamless EV handover, EV revocation, and dynamic RSU & EV addition.
- Using informal security analysis and formal security analysis with the help of random oracle model and scyther tool, The EV-Auth can withstand well-known security attacks, and achieve user anonymity and untraceability which is demonstrated in this paper.
- The advantages of EV-Auth are highlighted by comparing with the existing protocols in terms of communication costs and computation time.

II. Related Work

Cryptographic authentication is a secure method of determining who they claim to be and obtaining consent to use the services and facilities. During the key establishment process, communicating entities in the network can compute and agree on a session key to secure subsequent communication over public channels.

Li *et al.* [12] demonstrated how to authenticate the identity and message integrity of an EV with the help of a service provider. The information generated by the EV is used by the service provider to detect fraudulent charging activities. The EV encrypts and transmits all charging information to RSUs. Rather than supplying a session key for each RSU, the same encryption key can be shared by multiple RSUs to reduce complexity. When an EV transmits a session key to the RSU, the RSU broadcasts it to all nearby RSUs. While EV authentication is insufficient in and of itself, an adversary may impersonate a RSU without further verification. As a result, network security requires authentication between EVs and RSUs. Informal discussion on how the protocol protects against several well-known security attacks is also provided.

Hussain *et al.* [13] developed a privacy-aware mutual authentication protocol for EVs and CPs that enables the Charging Service Providing Authority (CSPA) to choose between two authentication mechanisms: direct authentication or hash chain-based authentication. The conventional vehicular ad hoc network (VANET) is composed of EVs equipped with an onboard device to communicate with the charging system. The VANET authorities can handle registration and revocation of charging devices. Each VANET body must obtain a certificate from the motor vehicle department. The CSPA is responsible for maintaining and powering the CPs. The authors have put efforts to develop mutual authentication and demonstrated that their protocol prevents impersonation attacks, replay attacks, and conflicts between electric vehicles and charging systems. However, security concerns such as man-in-the-middle attacks, denial-of-service attacks, and integrity have not been investigated.

Li *et al.* [14] asserted that digital signatures, one-time signatures, and authentication challenge-responses are still unsuitable for use in EV dynamic charging systems due to the computational complexity. However, the authors have used symmetric key encryption and pseudonyms to protect the privacy of an EV. The pad owner (PO) assigns pseudonyms and a secret key set to CPs. The protocol's functionality has been evaluated on a Raspberry Pi Model B using the Crypto++ 5.6.2 library. However, the outcomes indicate that the proposed approach is extremely successful, as vehicle verification takes only 0.11 ms. To avoid free riders, their model requires a certain degree of separation between charging pads. Although, this configuration degrades the device's charging performance. Heavy trucks that require more energy to drive will gradually be allowed to enter the lane.

Rabieh *et al.* [15] developed a privacy-aware protocol that protects drivers' privacy while autonomously authenticating EVs and charging stations. The VANET-based architecture is composed of the following entities: a charging company that is responsible for constructing the infrastructure necessary to enable EV charging; and a charging management centre that monitors and manages charging pads and charging controllers. Along the roadway, charging controllers are installed and connected to the charging management centre via rapid communication technologies. The MIRACL crypto library performs an evaluation of the computing times required for cryptographic operations. Additionally, they demonstrated how their system enables authentication between communicating principles while maintaining privacy, anonymity, and unlinkability. However, they have not examined several attacks that could have a system-wide impact, such as impersonation attacks and injection attacks.

Roman *et al.* [16] proposed an authentication protocol for charging while driving-wireless power transfer charging systems in a cloud environment that requires customers to purchase tickets from a company charging service prior to accessing a charging station. They reasoned that the charging server of the company would be capable of monitoring and controlling a large number of charging stations. The EVs communicate with FS, RSUs, CPs, and a corporate charging service in order to purchase tickets and charge. The AVISPA tool is used to rigorously verify the security of protocols. The system ensures message integrity, EV privacy, and mutual authentication among communicating entities. Their protocol, on the other hand, was vulnerable to a DoS attack and lacked adequate mutual authentication.

Hamouid *et al.* [17] presented a new privacy-aware protocol for electric vehicle dynamic charging systems that conceals the

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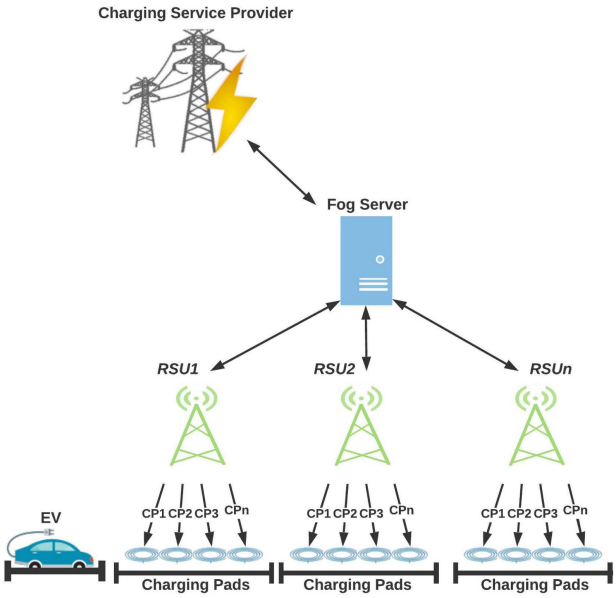


Fig. 1. Proposed Network Model.

EV's geolocation during the charging process. Additionally, their protocol is notable for its anonymity, rapid authentication between EV and CP via a single message exchange without the involvement of a CSP, and prevention of EV double-spending. The system is composed of four components: a bank, a CSP, an RSU, and a CP. It has been demonstrated the protocol's efficiency in terms of communication and computing costs by comparing it to other pertinent protocols.

Gunukula *et al.* [18] proposed a protocol in which EVs must first purchase anonymous coins from a bank before accessing a charging service provider in order to ensure anonymous expenditure and verification. Due to the coins created with partial blind signatures, it is difficult to identify a specific EV. The network model is comprised of a bank that is responsible for both the sale and validation of coins. A charging service provider manages the RSUs group in the charging station. The RSU is responsible for a network of charging stations beneath the highway's surface. The charging pads are in charge of supplying electricity to the electric vehicle. Attacks on the drivers' and payers' privacy are easily avoidable. They discussed payment-related attacks, including "dual-spending" and "man-in-the-middle" attacks.

According to the analysis performed on related protocols, the vast majority of them fall short of security objectives. Moreover, dynamic addition of EVs and RSUs, and the handover EVs among RSUs are not addressed. Indeed, several protocols lack formal security analysis excluding Roman *et al.* [16]. Therefore, an authentication protocol suite for EV dynamic charging systems that enables seamless EV handover between RSUs is proposed in this paper.

III. Network Model

The proposed network model is illustrated in Fig. 1, and it includes entities such as a charging service provider, a fog server, roadside units, and charging pads, each of which has the following roles and responsibilities.

- *Charging Service Provider:* CSP appears to be prepared to make an investment in dynamic charging stations. It would be in charge of enforcing the required infrastructure to charge EVs same as RSUs and CPs. It is indeed a trusted offline entity that handles everything essential to set up a dynamic charging station.
- *Fog Server:* The FS is the primary entity in the dynamic charging system, as it maintains complete control of all RSUs. Through an internet connection, the FS communicates with all RSUs. It is in charge of EV and RSU registration via the transport layer security protocol [19].
- *Road Side Units:* RSUs are access points located throughout the charging area of the road, which can span between 300 and 1000 metres. Each RSU is capable of communicating with and managing a large number of charging pads. They outperform electric vehicles in terms of computing and storage capacity. An RSU stores its secret values in tamper-resistant hardware and verifies/computes the messages throughout the charging process [20].
- *Electric Vehicle:* During the registration phase, an EV is equipped with an OBU capable of performing the required tasks. Additionally, an OBU is a tamper-resistant device equipped with a small memory for storing sensitive data securely. EVs can communicate with both RSUs and CPs via dedicated short-range communications (DSRC).
- *Charging Pads:* The CPs are the components that enable electric vehicles to be charged. It is intended to assign a set of charging pads to a specific RSU. Following verification, each CP injects energy into the EV. Each CP is capable of charging an EV independently of the others [21].

EV-Auth presume that RSUs are spaced enough apart to avoid overlapping zones between any two RSUs. In practice, it is difficult to eliminate completely the overlapping zones between any two RSUs. As a result, charging pads are not inserted in the overlapping zones. Notably, this kind of charging may increase charging efficiency by removing the requirement for charging pads in overlapping zones to be powered concurrently by two RSUs for the same automobile.Additionally, as illustrated in Fig. 5, the EV-Auth enables EVs to roam between multiple RSUs. When an EV wishes to change RSU regions, it sends a request to the current RSU, which notifies the subsequent RSU to ensure uninterrupted charging. The target RSU validates the EV before allowing it to charge. The term "handover authentication" is a blanket term that refers to the entire process.

IV. Threat Model

The primary goal of an adversary is to get unauthorized access to and benefit from the network's resources. An adversary might seek to get sensitive information by executing passive or active assaults on communications exchanged across a public channel between communicative parties. In a computer or networked system, an adversary model is a formalization of an attacker. Depending on how comprehensive this formalization is, the adversary might be an algorithm or a set of assertions about skills and intentions [22]. In this paper, the first and most widely used adversary models, i.e., the Dolev-Yao model [23], CK-model [24] are adopted, where the adversary can listen to

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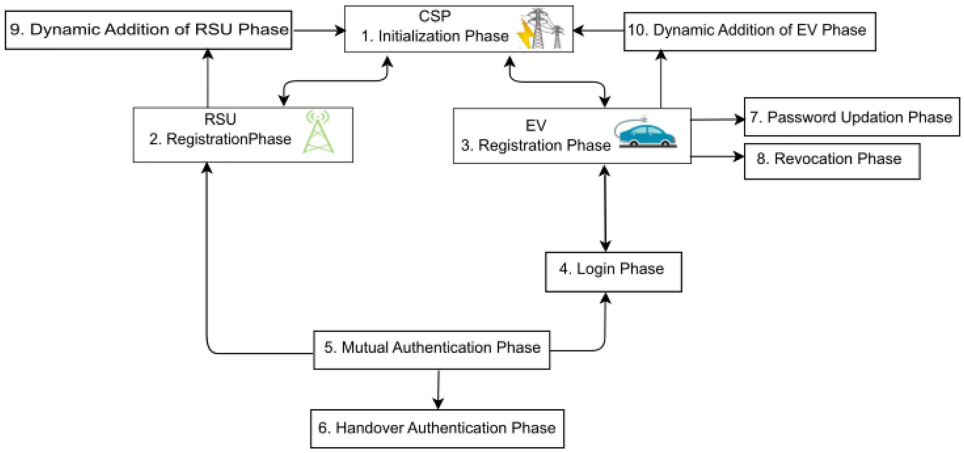


Fig. 2. Flow process of the EV-Auth protocol.

all traffic on a network. In addition, the following security and performance considerations are taken into account:

- To limit/control the damage in the event of an attack, unique keys for each individual RSU are used.
- A tamper-proof device is a small, secure storage area within an OBU used to store secret values. A Tamper Proof Device cannot be tampered, as it contains sensors that detect hardware tampering, and if this occurs, all stored values are immediately deleted from the memory [25].
- Even if the proposed system entities collide with each other, it is impossible to determine the true identity of an EV.

V. EV-AUTH

The EV-Auth consists of the following phases: *A)* Initialization Phase, *B)* RSU Registration Phase, *C)* EV Registration Phase, *D)* Login Phase, *E)* Mutual Authentication Phase, *F)* Handover Authentication Phase, *G)* EV Password Updation Phase, *H)* EV Revocation Phase, *I)* Dynamic Addition of RSU Phase, and *J)* Dynamic Addition of EV Phase. The CSP and FS are collocated with good coordination among them. The notations used in EV-Auth are shown in Table I. The flow process diagram of the all the phases of EV-Auth are depicted in Fig. 2.

A. Initialization Phase

The *CSP* generates the required system parameters as follows:

Step 1: *CSP* selects collision-resistant one-way cryptographic hash function *H*(.).

Step 2: *CSP* picks long-term private key SK_{RSU_i}, and keeps secret.

Step 3: *CSP* selects G_{pad} as a group key and delivers G_{pad} to *RSU* and *CP*s.

B. RSU Registration Phase

The RSU_i that plays a key role in the charging system can register with the *CSP* through secure channel.

Step 1: RSU_i picks an identifier ID_{RSU_i} and sends registration request ID_{RSU_i} to the *FS*.

TABLE I
NOTATIONS

Symbol	Description
ID_{EV}	Identity of EV
PW_{EV}	Password of EV
ID_{RSU_i}	Identity of ith RSU
SK_{RSU_i}	Long-term secret key of ith RSU
G_{pad}	Group key shared by RSU and CP
PU_{RSU_i}, PR_{RSU_i}	Public and Private key pair of i^{th}RSU
R_{EV}, n_1, n_2, n_3, X_i, K_i	Random numbers
Tag_i	i^{th} Tag computed using message authentication code (MAC)
$H(.)$	One-way hash function
$\|$	Concatenation
$\oplus$	Exclusive-OR operation
Æ	Adversary
SK	Session key
CSP	Charging service provider
FS	Fog server
RSU	Road side unit
EV	Electric vehicle
CP	Charging pad

Step 2: *FS* computes $K_{RSU_i} = H(ID_{RSU_i} \parallel SK_{RSU_i})$ and stores ID_{RSU_i}, K_{RSU_i} in table T_{RSU}.

Step 3: *FS* chooses public and private keys PU_{RSU_i} and PR_{RSU_i} for each RSU_i, stores and delivers them along with $< K_{RSU_i}, H(SK_{RSU_i}) >$.

C. EV Registration Phase

This phase provides registration to the owners of *EV*s, who want to charge their vehicles at the charging station via secure channel.

Step 1: *EV* selects an identifier ID_{EV}, password PW_{EV}, and picks a random number $R_{EV} \in Z_p^*$. Then *EV* computes $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, $PWD_{EV} = H(PW_{EV} \parallel R_{EV})$, $Y_{EV} = R_{EV} \oplus H(ID_{EV} \parallel PW_{EV})$ and sends $< PID_{EV}, PWD_{EV} >$ to *FS*.

Step 2: *FS* checks for any duplication of PID_{EV} and then computes $Q_S = H(PID_{EV} \parallel K_{RSU_i})$, $R_S =$

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$Q_S \oplus PWD_{EV}$, $W_S = H(PID_{EV} \parallel SK_{RSU_i})$, and $C_{EV} = H(PID_{EV} \| W_S)$, keeps $< ID_{RSUi}, R_S >$ in table T_{EV}. *FS* also stores $< PID_{EV}, C_{EV}, T_{R=1} >$ in table T_C. Finally, delivers $< W_S, T_{EV}, H(SK_{RSU_i}) >$ to *EV*.

Step 3: *EV* computes $X_S = W_S \oplus PWD_{EV}$, $C_{EV} = H(ID_{EV} \parallel W_S)$ and replaces W_S with X_S and stores $< X_S, Y_{EV}, T_{EV}, C_{EV}, H(SK_{RSU_i}) >$ in a tamper-proof onboard memory.

D. Login Phase

When an *EV* wishes to charge at a charging station, the owner of the *EV* can initiate the login request by providing ID_{EV}, PW_{EV} as follows.

Step 1: *EV* computes $R_{EV} = Y_{EV} \oplus H(ID_{EV} \parallel PW_{EV})$, $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, $PWD_{EV} = H(PW_{EV} \parallel R_{EV})$, $W_S = X_S \oplus PWD_{EV}$ and then verifies whether $C_{EV} = H(PID_{EV} \parallel W_S)$ holds, *EV* broadcasts the request signal to the nearest *RSU* and obtains its identifier, then retrieves R_S from T_{EV}.

Step 2: *EV* extracts $Q_S = R_S \oplus PWD_{EV}$, chooses random number $n_1 \in Z_p^*$ and computes $B_{ES} = PID_{EV} \oplus H(ID_{RSU_i} \parallel n_1 \parallel H(SK_{RSU_i})$ and $D_{ES} = H(PID_{EV} \parallel Q_S \parallel n_1)$.

Step 3: Finally, *EV* sends login request message $< M_1 = B_{ES}, D_{ES}, n_1 >$ to RSU_i.

E. Mutual Authentication Phase

Step 1: RSU_i computes $PID_{EV} = B_{ES} \oplus H(ID_{RSU_i} \parallel n_1 \parallel H(SK_{RSU_i}))$, and then verifies whether $D_{ES} = H(PID_{EV} \parallel Q_S \parallel n_1)$ holds, then RSU_i authenticates the *EV*, otherwise the process terminates.

Step 2: RSU_i selects random number $n_2 \in Z_p^*$, computes $SK = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S)$ and $E_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$, sends $< M_2 = E_{ES}, n_2 >$ to *EV*.

Step 3: *EV* computes $SK = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S)$ and verifies whether $E_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$ holds, then *EV* authenticates RSU_i, otherwise process terminates. *EV* computes $F_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$ and sends $< M_3 = F_{ES} >$ to RSU_i.

Step 4: RSU_i verifies $F_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$ and confirms the authenticity of *EV* once again. Now the *EV* and RSU_i use computed key *SK* for further communications.

Step 5: Now, *EV* sends a charging request message $M_4 =< chargingrequest >$ to RSU_i.

Step 6: RSU_i picks X_i, K_i for *i = 1 to n* based on *EV* request, and sends message $M_5 =< E_{SK}(Xi, Ki) >$ to *EV*.

Step 7: Similarly, RSU_i compute $Tag_i = MAC(X_i, K_i)$ for all *i = 1 to n* and compute value $K_{Tag} = \sum_{i=1}^{n} Tag_i$. Sends message $M_6 =< E_{Gpad}(K_{Tag}) >$ to the first *CP* in the corresponding sector.Then first *CP* decrypts M_6, and keeps K_{Tag}.

Step 8: *EV* decrypts message M_5, and compute $Tag_i = MAC(X_i, K_i)$ for all *i = 1 to n* and compute value $K_{Tag} = \sum_{i=1}^{n} Tag_i$.

Step 9: Finally, *EV* sends computed $M_7 =< K_{Tag} >$ to *CP*, Then *CP* checks the validity of K_{Tag} by comparing it with stored K_{Tag}. If verification successful, then *EV* gets charged by passing through the n number of charging pads in the sector. The entire login and mutual authentication process of EV-Auth is shown in Fig. 3 and Fig. 4.

F. Handover Authentication Phase

The *EV* with identity PID_{EV} under the vicinity of RSU_i coverage area wants to move into the coverage area of subsequent RSU_j with identity ID_{RSU_j}, then RSU_j authenticates *EV* and establishes a session key for secure communication between them as depicted in Fig. 5.

Step 1: Initially, *EV* sends the handover request message to the current RSU_i, then RSU_i checks for the nearby *RSU* based on the *EV* moving direction and sends message $M_1 =< n_3, ID_{RSU_j} >$ to *EV*.

Step 2: Parallelly, RSU_i prepares a message $M_2 =< E_{PU_{RSU_j}}(PID_{EV}, n_3, ID_{RSU_i}) >$ and send it to subsequent RSU_j.

Step 3: *EV* checks the ID_{RSU_j} in table T_{EV}, then computes $G_{ES} = H(PID_{EV} \parallel ID_{RSU_i} \parallel n_3 \parallel ID_{RSU_j})$ and sends it to target RSU_j.

Step 4: RSU_j decrypts message M_2, and verifies whether $G_{ES} = H(PID_{EV} \parallel ID_{RSU_i} \parallel n_3 \parallel ID_{RSU_j})$ holds, then RSU_j authenticates *EV*, otherwise process terminates. RSU_j computes temporary session key $SK_T = H(PID_{EV} \parallel n_3)$ and picks X_i, K_i for *i = 1 to n* based on *EV* request, and sends message $M_3 =< E_{SK_T}(Xi, Ki) >$ to *EV*.

Step 5: Similarly, RSU_j computes $Tag_i = MAC(X_i, K_i)$ for all *i = 1 to n* and computes value $K_{Tag} = \sum_{i=1}^{n} Tag_i$. Sends message $M_4 =< E_{Gpad}(K_{Tag}) >$ to the first *CP* in the corresponding sector.Then first *CP* decrypts M_4, and keeps K_{Tag}.

Step 6: *EV* decrypts message M_3, and compute $Tag_i = MAC(X_i, K_i)$ for all *i = 1 to n* and compute value $K_{Tag} = \sum_{i=1}^{n} Tag_i$.

Step 7: Finally, *EV* sends computed $M_5 =< K_{Tag} >$ to *CP*, Then *CP* checks the validity of K_{Tag} by comparing it with stored K_{Tag}. If verification successful, then *EV* gets charged by passing through the *n* number of charging pads in the sector.

G. EV Password Updation Phase

When the user of *EV* wants to update the password by inputting the ID_{EV} and PW_{EV}, the following procedure can help him/her without the involvement of *FS*.

Step 1: *EV* computes $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, $PWD_{EV} = H(PW_{EV} \parallel R_{EV})$, $W_S =$

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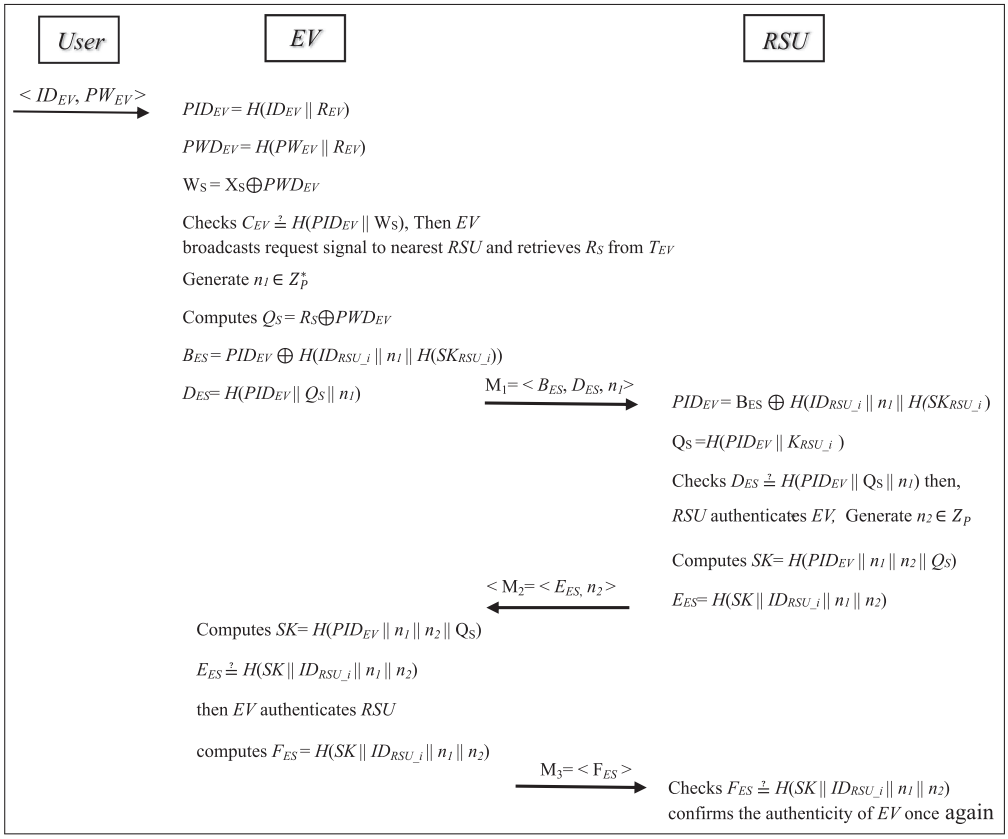


Fig. 3. Login and Mutual Authentication Phase (1).

$X_S \oplus PWD_{EV}$ and then verifies whether $C_{EV} = H(ID_{EV} \parallel W_S)$ holds, then login request is permitted.

Step 2: *EV* retrieves $R_S = W_S \oplus PWD_{EV}$ for all ID_{RSU} using the current password PW_{EV}. *EV* chooses a new password PW^*_{EV} and computes $PWD^*_{EV} = H(PW^*_{EV} \parallel R_{EV})$, $X^*_S = W_S \oplus PWD^*_{EV}$, $R^*_S = Q_S \oplus PWD^*_{EV}$ for all ID_{RSU}.

Step 3: *EV* updates table T^*_{EV} and parameters X^*_S, the *EV* contains $< X^*_S, T^*_{EV}, C_{EV}, H(SK_{RSU_i}) >$ in a tamper-proof onboard memory.

H. EV Revocation Phase

If the *EV* wants to revoke the available services, the following procedure can be performed over a secure channel after the successful login phase.

During the revocation phase, the *EV* will have to prove legitimacy and submit all credentials to *FS*. Upon receiving the request, *FS* sets $T_R = 0$ and updates (PID_{EV}, $C_{EV}, T_R = 0$) in table T_C, where $T_R = 0$ means *EV* revoked and inactive. When *EV* wants to resume services, then *EV* must prove $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, $PWD_{EV} = H(PW_{EV} \parallel R_{EV})$ by retrieving $Q_S = R_S \oplus PWD_{EV}$ using recent ID_{EV} and PW_{EV}. If the *EV* holds correct credentials, then *FS* resumes *EV* services and updates $< PID_{EV}, C_{EV}, T_R = 1 >$ in table T_C.

I. Dynamic Addition of RSU Phase

In this phase, a new road side unit RSU^{new} can become a member of charging system by sending a registration request to the *FS*. The new *RSU*'s information will be forwarded to all the registered *EV*s and *RSU*s periodically using their stored details. The RSU^{new} registration process is as follows:

Step 1: RSU^{new} sends registration request $< ID^{new}_{RSU_i} >$ to *FS*.

Step 2: *FS* computes $K^{new}_{RSU_i} = H(ID^{new}_{RSU_i} \parallel SK^{new}_{RSU_i})$ and stores $< ID^{new}_{RSU_i}, K^{new}_{RSU_i} >$ in table T_{RSU}.

Step 3: *FS* chooses public and private keys $PK^{new}_{RSU_i}$ and $SK^{new}_{RSU_i}$ for each RSU_i, stores and delivers them along with $< K^{new}_{RSU_i} >$.

Step 4: FS retrieves the $< PID_{EV}, C_{EV} >$ of all *EV*s and computes $Q^{new}_S = H(PID_{EV} \parallel K^{new}_{RSU_i})$, and then delivers it to *EV*s via secure channel. *FS* updates $< PID_{EV}, C_{EV} >$ in T_C. *EV* computes $R^{new}_S = Q^{new}_S \oplus PID_{EV}$ and stores it in T_{EV}.

J. Dynamic Addition of EV Phase

This phase provides registration to the owners of new *EV*, who wants to charge their vehicles at the charging station

Step 1: EV^{new} selects an identifier ID^{new}_{EV}, password PW^{new}_{EV}, and picks a random number $R^{new}_{EV} \in Z^*_p$. Then EV^{new} computes $PID^{new}_{EV} = H(ID^{new}_{EV} \parallel$

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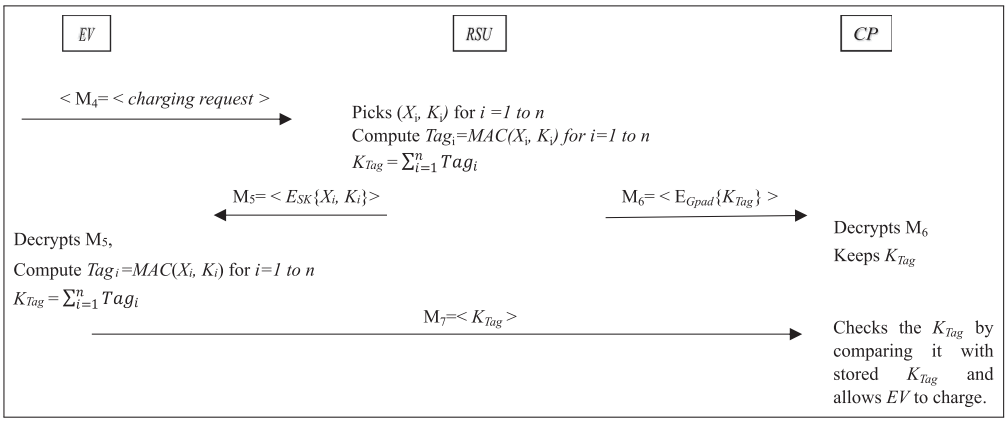


Fig. 4. Login and Mutual Authentication Phase (2).

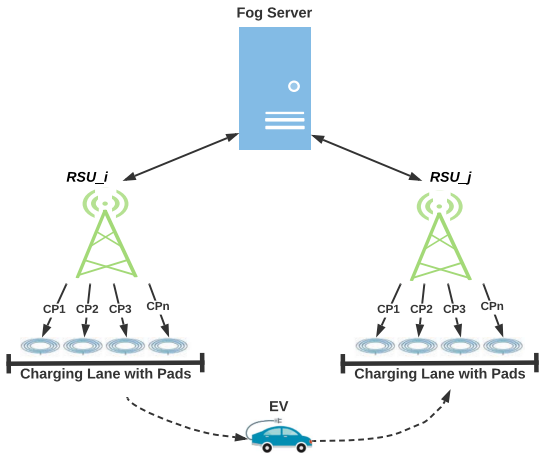


Fig. 5. EV Handover between RSUs.

R_{EV}^{new}), $PWD_{EV}^{new} = H(PW_{EV}^{new} \parallel R_{EV}^{new})$ and sends $< PID_{EV}^{new}, PWD_{EV}^{new} >$ to *FS*.

Step 2: *FS* checks for any duplication of PID_{EV}^{new} and then computes $Q_S^{new} = H(PID_{EV}^{new} \parallel K_{RSU_i})$, $R_S = Q_S^{new} \oplus PID_{EV}^{new}$, and $W_S = H(PID_{EV} \parallel SK_{RSU_i})$, keeps $< ID_{RSU_i}, R_S >$ in table T_{EV}. *FS* also stores $< PID_{EV}, C_{EV}, T_{R=1} >$ in table T_C. Finally, delivers $< W_S, T_{EV}, H(SK_{RSU_i} >$ to *EV*.

Step 3: *EV* computes $X_S = W_S \oplus PID_{EV}^{new}$, $C_{EV}^{new} = H(ID_{EV}^{new} \parallel W_S)$ and *EV* replaces W_S with X_S and stores $< X_S, T_{EV}, C_{EV}, H(SK_{RSU_i} >$ in a tamper-proof onboard memory.

VI. Security Analysis

This section conducts a formal security analysis of the proposed protocol suite, utilizing the random oracle model and the scyther tool, as well as an informal analysis, covering all propositions related to prominent attacks.

A. Formal Security Analysis Using Random Oracle Model

In this section, a formal analysis for our authenticated key agreement protocol in the random oracle model (ROM) [26] is provided. The benign parties involved in the protocol are users, roadside units (RSUs) and electric vehicles (EVs). The users and EVs share a low entropy password, and RSUs and EVs share a long-term, pair-wise secret key ($H(SK_{RSU_i})$). The objective of the protocol is to agree on a common session key (SK) based on the long-term secrets. A session of the protocol according to the description of the protocol is known as the instance of the protocol, and it is represented by $\mathcal{P}$. A party belonging to an instance is denoted by $\mathcal{G}$. Several instances of the protocol form oracle of the protocol, (1, 2, 3,..., $\mathcal{O}^{\mathcal{P}_\mathcal{G}}$. In such oracle ($\mathcal{O}^{\mathcal{P}_\mathcal{G}}$), i^{th} instance of the user, EV and RSU is represented by $User$, EV, and RSU.

Now, we introduce ability queries of the adversary ($\mathcal{A}$) and some notions and definitions to prove the security of our protocol. The ability of the adversary is modelled by the following queries.

Send (User, EV and RSU): The adversarial ability of message initiation and sending messages as per the protocol is modelled using this query. Specifically, Send (User) initiates the protocol, and Send (EV) and Send (RSU) models the message exchange behaviour of electric vehicles and RSUs, correspondingly.

Execute (User, EV and RSU): This query models the adversarial ability to eavesdrop on the communications channel when honest parties exchange messages in the communication channel. For example, the adversary issues Execute (EV) to overhear the communications made by electric vehicles in the communication channel.

Hash (M, $\sigma(M, K)$) [ψ(M, $\sigma(M, K)$)]: The message authentication code algorithm used in the protocol is modelled as the keyed hash table ψ, where M is the input and K is the secret key. The table consists of three columns, namely name of the party, input and tag (message authentication code). When the adversary issues the hash query to the table for an input, the table checks the input column. If the inout is already queried by the honest parties, then it gives the corresponding tag to the adversary. If the input is not found in the table, then the table picks a random number. The picked random number is inserted in the column of the tag, and the random number is given as the output to the adversary for the queried input.

Test ($\mathcal{P}$): Compared to the previous queries, this query is issued to an arbitrary instance of the protocol by the adversary. Also, this query has to be issued only once by the adversary.

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Once the test query is issued to an arbitrary instance, the instance is called the targeted session or the Test session. But, before issuing the Test query, the adversary commits to a bit (b). When the query is issued to the Test session that has a bit ($b^{'}$) chosen by honest parties of the instance, the query returns the session key of the instance if $b = b^{'}$; otherwise, the query returns a random string in the size of the session key to the adversary.

Session-state reveal (U, EV and RSU): The internal state of users, electric vehicles and roadside infrastructures can be learnt by the adversary using this query. An example of the internal state of the protocol is ephemeral values that are used in the protocol. The adversary issues this query to all sessions except the Test session.

Session-key reveal (U, EV and RSU): The adversarial ability to learn the past session keys is modelled in this query. The adversary issues this query to $\mathcal{O}^{\mathcal{P}}$ except the Test session. Upon issuing this query to $\mathcal{O}^{\mathcal{P}}$, the query brings all previously established session keys with the party to the adversary.

Corrupt (U, EV and RSU): The adversarial ability to acquire long-term secret (password) is modelled in this query. If the adversary issues this query to $\mathcal{O}^{\mathcal{P}}$ except the Test session, the query returns the password of $\mathcal{P}_{\mathcal{G}}$ to the adversary. Specifically, this query brings the password of the user to the adversary.

Remark 1: The hash $H(\cdot)$ oracle is instances of the random oracle (RO).

The following definitions and assumptions are introduced to prove the security of our authenticated key agreement protocol.

Definition 1 (Accepted and sid [27]): An instance of $\mathcal{P}$ is *accepted* if all the steps of the protocol have proceeded exactly (to the protocol specification) until the last step. The concatenation of the messages exchanged in $\mathcal{P}$ forms the session identifier (*sid*) of the instance.

Definition 2 (Partnership [28]): The protocol instance $\mathcal{P}$ is run by three parties $User$, EV and RSU and the parties are called *partners* if the following two conditions hold:

Step 1: $User$, EV and RSU are running the i-th session and

Step 2: the session is accepted.

Definition 3 (Freshness): An instance is said to be fresh if all the following conditions hold: 1) The protocol satisfies the accepted state. 2) The identical session identity (sid) is held by benign participants of the protocol due to the proper execution of the instance of the protocol.

For the targeted instance, the adversary can issue the following queries alone: 1) Session-state reveal query ($\neg$ Test $\mathcal{P}$) and Session-key reveal query ($\neg$ Test $\mathcal{P}$), 2) Corrupt query ($\neg$ Test $\mathcal{P}$). It is noted that the aforementioned two queries cannot be asked together by the adversary.

Definition 4 (Negligible function [29]): A function $\omega(n)$ is *negligible*, if for all $c > 0$, there exists $n_0 \in \mathbb{Z}^+$ such that

$$|\omega(n)| \leq \frac{1}{n^c} \text{ for all } n \geq n_0.$$

Definition 5 (One-wayness property [30]): Given a hash output H(x) of length n bits, it is computationally infeasible to find x in polynomial time.

Assumption 1 (One-wayness assumption): Any probabilistic polynomial time adversary against the one-wayness property of the hash function has a negligible advantage in finding the integer x (input) for the given H(x) (output) of size n bits

$$Adv_{\mathcal{A}}^{OW}(n) = Pr[\mathcal{A}(1^n, H(x))] = x] \leq \omega(n).$$

Definition 6 (Weak collision resistance property [30]): Given an input and a hash output (x_1, H(x_1) with H(x_1) of length n bits, it is computationally infeasible to find x_2 in polynomial time such that H(x_1) = H(x_2) and $x_1 \neq x_2$.

Assumption 2 (Weak collision resistance assumption): Any probabilistic polynomial time adversary against the weak collision resistance property of the hash function has a negligible advantage in finding the integer x_2 for the given x_1 and $H(x_1)$ of size n bits

$$Adv_{\mathcal{A}}^{WC}(n) = Pr[\mathcal{A}(1^n, x_1, H(x_1) = H(x_2))] = x_2] \leq \omega(n).$$

Definition 7 (Semantic security): The semantic security of our authenticated key exchange protocol suite is defined as the negligible advantage gained by the adversary for acquiring the session key in the following several stages of the security experiment. Stage 1 (initialisation): The adversary is initialised with the oracle of the protocol after choosing a bit ($b^{'}$) for every session that is fresh. Stage 2 (adversarial access): The adversary issues a polynomial number of queries to the following oracles: Send, Execute and Hash. Stage 3 (adversarial attack): After collecting the polynomial number of queries in the previous stage, the adversary commits to a bit (b) and issues the Test query to an arbitrary session that is chosen by benign parties for which the freshness holds. Stage 4 (adaptive attack of the adversary): The adversary continues its attack using its ability query without losing the freshness of the targeted session. Stage 5 (adversarial output): The adversarial bit is compared with the bit of the targeted session. If the bits are matched, then the adversary holds the session key of the targeted session; otherwise, it holds a random number of size n bits for the session key. The protocol is semantically secure if the adversarial advantage is negligible or

$$Adv_{\mathcal{A}}(\mathcal{P}, n) = Pr[\mathcal{A}(b^{'} = b] - \frac{1}{2} \leq \omega(n).$$

It is noted that the semantic security is defined similar to the semantic security definition in [31].

Definition 8 (Difference lemma [32]): Let P, Q and F be events defined in some probability distribution, and suppose that $P \wedge \neg F \equiv Q \wedge \neg F$. Then, $Pr[P] - Pr[Q] \leq Pr[F]$.

Theorem 1: Theorem 1. Given $\mathcal{P}$, the following assumptions hold:

- a session key of size n bits,
- a dictionary of size $|\mathcal{D}|$,
- a random number of size m bits,
- $q_{SS} = 4 \times (q_s + q_e)$ be the total number of sessions,
- a long term secret key can be guessed with an advantage Adv^{sk}. The one-wayness property of the message authentication code can be broken with the advantage Adv^{OW}. The weak collision resistance property of the message authentication code can be broken with the advantage Adv^{WC},

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- the hash function generates MACs with ℓ bits,
- $\mathcal{A}$ has access to all public oracles and can issue a polynomial number of queries all public oracles.

Then semantic security of the protocol can be broken by $\mathcal{A}$ with the advantage

$$\begin{aligned} Adv_{\mathcal{P}}(n) \leq & \max\left\{\frac{q_h^2+4(q_s+q_e)^2}{2^\ell}, Adv^{WC}+\frac{4(q_s+q_e)^2}{2^\ell}\right\} \\ & + \frac{Q_{SS}^2}{2^{m-1}} + \max\left\{Adv^{OW}, \frac{1}{2^m} + Adv^{sk}\right\} \\ & + \max\left\{\frac{1}{2^m}, \frac{(q_s+q_d)^2}{|\mathcal{D}|} + \frac{1}{2^m}\right\} \\ & + \max\left\{\frac{1}{2^m} + \frac{(q_s+q_d)^2}{|\mathcal{D}|}, \frac{1}{2^m} + Adv^{sk}\right\} + \frac{Q_{SS}}{2^n} \end{aligned}$$

where q_h, q_s, q_e and q_d stands for the number of queries to the hash, send, execute and dictionary oracles.

Outline of the proof: We define six games starting from G_0 to G_5 to prove the semantic security of the protocol. The starting game G_0 corresponds to an attack against the real protocol. The second game G_1 is similar to G_0 except $\mathcal{A}$ finds collisions in ψ. The third game G_2 is the same as G_1 except that the event repeat occurs when EV and RSU choose the same random number (n_1 and n_2). The fourth game G_3 presents the known key secrecy of the protocol. The fifth game G_4 presents the online and offline dictionary attacks against the protocol. The final game G_5 presents the implication to the protocol in the case of adaptive adversarial attack by the adversary.

The main idea of the proof is to depict that the adversary begins with some initial advantage of winning the starting game. Then, using the difference lemma, it can be claimed that the adversary does not gain sufficient advantage in the subsequent games until the final game. Moreover, in the final game, it can be observed that the advantage of the adversary is zero in acquiring the session key of the targeted session. Clearly, the adversarial advantage for getting the session key is merely the advantage involved in guessing a random number.

Proof: We defined six games starting from G_0 to G_5. E_i is the event that $\mathcal{A}$ wins the game G_i, where $i = 0, 1, 2, 3, 4, 5$.

Game G_0: This game is an attack against the real protocol. As the adversary has access to the public oracles, it collects a polynomial number of queries from the public oracles. Then, it commits to a bit (b) and issues the Test query to an arbitrary session that is chosen by $\mathcal{P}_{EV}$. which has the bit ($b^!$).

The winning probability of the adversary as per the semantic security definition is given by

$$Adv_{\mathcal{A}}(\mathcal{P}, n) = Pr[\mathcal{A}(b^! = b] - \frac{1}{2}$$

Game G_1: This game is similar to the previous game except that this game fails due to the occurrence of collision in inputs of ψ. The collision can happen for four message authentication codes, namely B_{ES}, D_{ES}, E_{ES} and F_{ES}. If the collision happens for B_{ES}, then the adversary can create forgery for its identity. If a collision happens for D_{ES}, then the adversary can violate mutual authentication between EV and RSU by creating a forgery. If the adversary can find a collision for E_{ES}, then it can inject a fake session key to EV. Similarly, if the adversary can find a collision for F_{ES}, then the adversary can complete the key authentication step by a forgery. To forge the four tags, the adversary queries ψ independently. This means that the four message authentication codes are modelled as four independent random oracles. Then, according to the birthday paradox [33], the occurrence of the collision probability in each ψ is upper bound by $\frac{q_h^2}{2^\ell}$. Moreover, since the adversary initiates the protocol, the total number of send and execute queries are upper bound by $\frac{4(q_s+q_e)^2}{2^\ell}$. The adversary can also find weak collision (EUF-CMA) for the inputs with advantage $Adv^{WC}+\frac{4(q_s+q_e)^2}{2^\ell}$. Hence, the occurrence of the collision is upper bound by max $\{\frac{q_h^2+4(q_s+q_e)^2}{2^\ell}, Adv^{WC}+\frac{4(q_s+q_e)^2}{2^\ell}\}$. As the adversary wins this game with negligible probability, it is evident that the advantage gained by the adversary in this game is negligible in the security parameter. As per the difference lemma, we have

$$\begin{aligned} \mid Pr[E_1] - Pr[E_0] \mid \leq & \max\left\{\frac{q_h^2 + 4(q_s+q_e)^2}{2^\ell}, \right. \\ & \left. \times Adv^{WC} + \frac{4(q_s+q_e)^2}{2^\ell}\right\}. \end{aligned}$$

Game G_2: This game and the previous games are identical except that the simulation of the game fails for the occurrence of the repeat event. The event repeat occurs if n_1 and n_2 are chosen by another instance. If the game is not aborted, then the adversary can have a parallel session with the benign parties of the protocol. As the total number of sessions/instances is Q_{SS}, the failure probability is upper bound by $\frac{Q_{SS}^2}{2^{m-1}}$. Using the difference lemma, we present that the adversary does not gain any non-negligible advantage

$$\mid Pr[E_2] - Pr[E_1] \mid \leq \frac{Q_{SS}^2}{2^{m-1}}.$$

Game G_3: This game presents the known key secrecy of the protocol. This game resembles the previous game except that failure occurs in this game if the adversary finds the correct session key. In this game, the adversary, deploys its ability queries, namely the session-state reveal and session-key reveal to all sessions except the targeted session to acquire ephemeral secrets and session keys, respectively. Despite acquiring the ephemeral secrets and session keys of the remaining sessions, the adversary cannot acquire the session key of the targeted session because the session key is computed by H($PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_s$). In H($PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_s$), n_1 and n_2 are public values, so the acquisition of previous session keys and ephemeral values, namely as H($ID_{RSU_i} \parallel n_1 \parallel$ H(SK_{RSU_i})) and Q_s do not help the adversary to compute the targeted session key. Nonetheless, the adversary can acquire the targeted session secrets in the following cases. Case i: From the acquired Q_s of the remaining sessions, the adversary can acquire $K_{RSU_i} = H(SK_{RSU_i})$ with the advantage Adv^{OW} for constructing a valid session key of the targeted session. Case ii: From the acquired SK of the remaining sessions, the adversary can acquire Q_s and PID_{EV} with the advantage Adv^{OW}. The previous two cases correspond to the advantage involved as the result of the deployment of the session state and session key queries, respectively. But, if the adversary

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knows R_{EV} and SK_{RSU_i}, then it can compute the Test session key. The advantage for guessing the R_{EV} and SK_{RSU_i} are upper bound by $\frac{1}{2^m}$ and Adv^{sk}, correspondingly. As per the difference lemma

$$\mid Pr[E_3] - Pr[E_2] \mid \leq max\left\{Adv^{OW}, \frac{1}{2^m} + Adv^{sk}\right\}.$$

Game G_4: This game presents the security of the session key of the protocol against the online and offline dictionary attacks for the password. This game is identical to the previous game. This game fails in the event of the adversarial acquisition of the session key of the targeted session due to the compromise of the password.

Case I (online dictionary attack): The adversary can acquire the password of the targeted session after issuing the q_d number of queries to $\mathcal{D}$ via the online dictionary attack. It is noted that the passwords are stored in a hashed format. So, in the online dictionary attack, the adversary acquires the password with the advantage $\frac{(q_s+q_d)^2}{|\mathcal{D}|}$. The adversary needs R_{EV} although the password is obtained through the online dictionary attack. The total advantage is upper bound by $\frac{(q_s+q_d)^2}{|\mathcal{D}|} + \frac{1}{2^m}$ since $\frac{1}{2^m}$ is the probability required for $\mathcal{A}$ to predict R_{EV}.

Case II (offline dictionary attack): In this attack, the adversary issues the corrupt query to $\mathcal{P}$ except for the test session and acquires the password of EV. Acquiring the password of remaining sessions cannot help the adversary to compute PWD_{EV} of the targeted session since R_{EV} is unknown to the adversary. But, having the password, R_{EV} and SK_RSU_i, the adversary can compute the session key of the targeted session. The R_{EV} can be acquired by the adversary with the probability $\frac{1}{2^m}$. SK_{RSU_i} can be acquired by the adversary with advantage Adv^{sk}, As per difference lemma, we have

$$\mid Pr[E_4] - Pr[E_3] \mid \leq \max\left\{\frac{1}{2^m} + \frac{(q_s+q_d)^2}{|\mathcal{D}|}, \frac{1}{2^m} + Adv^{sk}\right\}.$$

Game G_5: This game presents the implication of the adaptive adversarial attack in the protocol, and it is similar to the previous game. Usually, after the formation of $\mathcal{O}^{\mathcal{P}}$, the adversary is given a targeted session to evaluate the security of the session key. The adversary is allowed to issue session-state and session-key, and corrupt queries to remaining sessions except for the test session. But, if the adversary correctly guesses the test instance from $\mathcal{O}^{\mathcal{P}}$, then the adversary can commit to the bit $(b^!)$ of the test session. Finally, it can obtain the session key of the test session. Therefore, this game fails in the event of the adversarial guess of the test session. The total number of sessions in $\mathcal{O}^{\mathcal{P}}$ is Q_{SS}. The adversarial guess for the test session is upper bound by $\frac{1}{Q_{SS}}$. In the event that the adversary cannot guess the test session is upper bound by (1- $\frac{1}{Q_{SS}}$). According to the difference lemma, one can get

$$\mid Pr[E_5] - Pr[E_4] \mid \leq \frac{Q_{SS}}{2^n}.$$

In the final game, the winning probability of the adversary is negligible in the security parameter, n. Also, the adversarial probability of guessing the session key from the random number is $\frac{1}{2}$. Thus, the adversarial advantage of acquiring the session key is zero; or, it has the advantage of guessing a random number.

B. Informal Security Analysis

One can describe the proposed protocol's security analysis, outlining its security and privacy resilience against several critical vehicular communication attacks.

Proposition 1: The EV-Auth provides protection against EV impersonation attacks.

Proof: Case 1: To impersonate a legitimate EV, Æ must construct a login request message $M_1 =< B_{ES}, D_{ES}, n_1 >$. However, before initiating an authentication request, s/he should complete the login procedure. During login phase, EV computes $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, $PWD_{EV} = H(PW_{EV} \parallel R_{EV})$, $W_S = X_S \oplus PWD_{EV}$ and then validates whether the condition $C_{EV} = H(ID_{EV} \parallel W_S)$ holds. Without the necessary credentials, Æ will be unable to advance to the next stage. On the other hand, the likelihood of obtaining the correct ID_{EV} and PW_{EV} values is extremely remote.

Case 2: EV does not provide personally identifiable information to any RSU_i, which may acquire PID_{EV} of legitimate EV only during the login and mutual authentication phases by using $PID_{EV} = B_{ES} \oplus H(ID_{RSU_i} \parallel n_1 \parallel H(SK_{RSU_i})$. For instance, if any RSU_i transforms into Æ and wishes to emulate a legitimate EV, s/he still require K_{RSU_i} of the targeted RSU_i to construct $Q_S = H(PID_{EV} \parallel K_{RSU_i})$. Q_S value is unique for each RSU_i in the EV-Auth protocol, where $K_{RSU_i} = H(ID_{RSU_i} \parallel SK_{RSU_i})$. The restrictions listed above demonstrate how resilient our system is to EV impersonation attacks.

Proposition 2: The EV-Auth can withstand RSU impersonation attacks.

Proof: Consider a scenario in which a registered RSU_i disguised as Æ captures $< M_1 >$ and attempts to mimic a legitimate RSU_i by sending calculated message $< M_2^{Æ} >$ in response. When ID_{RSU_i} is assumed to be with Æ, then s/he can hardly obtain $PID_{EV} = B_{ES} \oplus H(ID_{RSU_i} \parallel n1 \parallel H(SK_{RSU_i})$ and n_1 from the seized M_1. The Q_S value of the targeted RSU_i required in order to compute the response $M_2^{Æ}$, since each RSU_i has unique long-term key K_{RSU_i} which is computed based on ID_{RSU_i} as $K_{RSU_i} = H(ID_{RSU_i} \parallel SK_{RSU_i})$. Thus, Æ cannot compute $SK = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S)$, $E_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$ and reply EV.

One can consider another scenario in which Æ sends message $M_2^{Æ} =< E_{ES}^{Æ}, n_2 >$ to EV after computing $SK^{Æ} = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S^{Æ})$, $E_{ES}^{Æ} = H(SK^{Æ} \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$. Upon receiving the response, EV computes $SK = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S)$, $E_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$ and can identify it as a malicious attempt due to the non-equivalence of messages $E_{ES}^{Æ} \neq E_{ES}$. The above assertions demonstrate that the EV-Auth protocol can resist RSU impersonation attacks.

Proposition 3: Given EV-Auth, mounting man-in-the-middle attack by adversary succeeds with negligible probability.

Proof: When data is transmitted over a shared communication channel and Æ comprehends the exchanged messages, a man-in-the-middle attack against the system is possible. The EV and RSU exchange messages $M_1 =< B_{ES}, D_{ES}, n_1 >$, $M2 =< E_{ES}, n_2 >$, $M_3 =< F_{ES} >$ over public channel during login and mutual authentication phases. Therefore, Æ can capture these values to understand messages exchanged between

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EV and RSU. According to the EV-Auth protocol, B_{ES}, D_{ES}, and E_{ES} are calculated as $B_{ES} = PID_{EV} \oplus H(ID_{RSU_i} \parallel n_1 \parallel H(SK_{RSU_i})$, $D_{ES} = H(PID_{EV} \parallel Q_S \parallel n_1)$, $E_{ES} = H(SK \parallel ID_{RSU_i} \parallel n_1 \parallel n_2)$ respectively using one-way hash function. Hence, Æ cannot find any vital information such as $Q_S = H(PID_{EV} \parallel K_{RSU_i})$, $SK = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S)$, which can be used to decipher the communication between the EV and the RSU, and these parameters are computed on a session-by-session basis. As a result, it can be highlighted that the proposed protocol is resistant to man-in-the-middle attacks.

Proposition 4: Given EV-Auth, any attempt of replay attack by the adversary succeeds with negligible probability.

Proof: Æ may attempt to establish a new session by replaying the previously sent message $< M_1 >$. RSU receives the message $< B_{ES}, D_{ES}, n_1 >$ and saves the pair (PID_{EV}, n_1) in its database during the login and mutual authentication phases. If Æ replays the same message $< M_1^{Æ} >$, RSU gets $(PID_{EV}, n_1^{Æ})$ and compares it to the stored (PID_{EV}, n_1). When RSU discovers $n_1^{Æ} \neq n_1$, it ignores the request and terminates the process. To avoid the replay attack on response messages $M_2 =< E_{ES}, n_2 >$ and $M_3 =< F_{ES} >$, both EV and RSU take a similar approach. Yet, the EV-Auth can withstand replay attacks with the adoption of random numbers n_1, n_2, n_3.

Proposition 5: The EV-Auth can withstand privileged insider attacks.

Proof: EV does not send the plain credentials or the credentials digest to the FS during the user registration phase. EV submits $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, $PWD_{EV} = H(PW_{EV} \parallel R_{EV})$ to FS, where $R_{EV} \in Z_P^*$ is a random number. As a result, an insider cannot obtain the original credentials of any EV. Additionally, the proposed protocol is based on the premise of not storing password verification tables, with entity authentication accomplished through the validation of incoming messages such as $D_{ES} = H(PID_{EV} \parallel Q_S \parallel n_1)$. This way, the EV-Auth prevents the insider attacks.

Proposition 6: The EV-Auth offers User anonymity and untraceability.

Proof: Vehicle users are genuinely uninterested in disclosing any personal information, such as user identification and relevant parameters, in order to reveal their actions and secret messages. If Æ is able to obtain private information about a registered vehicle user, the communication system is unable to maintain user anonymity to the benefit of registered vehicle users. In the EV-Auth protocol, both EV and RSU exchange messages $< M_1, M_2, M_3 >$ during the login and authentication phases. All these values are arbitrary for each session due to the random numbers n_1, n_2, and n_3. The EV's original identity ID_EV is encoded as $B_{ES} = PID_{EV} \oplus H(ID_{RSU_i} \parallel n_1 \parallel H(SK_{RSU_i})$, where $PID_{EV} = H(ID_{EV} \parallel R_{EV})$. All of these values are generated with the aid of a secure hash algorithm. As a result of their irreversible nature, it is not possible to derive any information from these computed values. In this way, the proposed protocol ensures user anonymity during the EV registration, login, and authentication phases. Additionally, the proposed protocol possesses an essential property known as untraceability. Due to the random nature of the parameters, it is unidentifiable and untraceable to the adversaries.

Proposition 7: The EV-Auth achieves forward secrecy.

Proof: It is a property of key agreement mechanisms to ensure that session keys cannot be computed in the future, even if long-term session key credentials are revealed. In the EV-Auth scheme, the session key is calculated as $SK = H(PID_{EV} \parallel n_1 \parallel n_2 \parallel Q_S)$. Since Æ knows n_1 and n_2, s/he needs other values (PID_{EV}, Q_S) for the session key computation. PID_{EV} is an pseudo identity of EV that may be calculated as $PID_{EV} = H(ID_{EV} \parallel R_{EV})$, where $R_{EV} \in Z_P^*$ is a random number. Q_S is a computed parameter using the formula $Q_S = H(PID_{EV} \parallel K_{RSU_i})$, where $K_{RSU_i} = H(ID_{RSU_i} \parallel SK_{RSU_i})$. In this case, Q_S and PID_{EV} are long-term variables of EV, but it is not feasible to obtain/compute these values anyhow due to unavailability of necessary values. Furthermore, the needed credentials (PID_{EV}, Q_S) are known only to EV and are not kept in memory. In each session key, the random numbers R_{EV}, n_1 and n_2 vary. Taking everything into account, the EV-Auth scheme satisfies the requirements for forward secrecy.

Proposition 8: The EV-Auth can withstand DoS attacks under certain circumstances.

Proof: It is used to degrade receiver performance by sending multiple messages with varying identities, anticipating a lengthy verification time and burden on the receiver's side. As a result, a vehicle may not receive critical information in a timely manner. The proposed protocol is built using low-cost operations (such as SHA-1, bit-wise XOR, and concatenation) that require minimal computing time. When a DoS attack is attempted, the receiver immediately checks its legitimacy and discards received messages if they are illegitimate. As a result, the proposed protocol is impervious to a DoS attack. Nonetheless, no assumption has been made about the possibility of committing DoS attacks via physical connection disruptions.

The EV-Auth scheme suite has ability to resist all well-known security features compared to other relevant security protocols for EV dynamic charging system as evident in Table II. Additionally, the EV-Auth scheme suite offers interesting feature called seamless handover which is beneficial for EV to roam over RSUs efficiently.

C. Formal Security Analysis Using Scyther Tool

Scyther [34], [35] offers protocol validation, presentation, analysis, specification, and generation. The Security Protocol Description Language (SPDL) defines roles as a series of events, where events representing information transfer and receipt. Initially, SPDL scripts are used to represent each location. The essential functions are those of the EV and the RSU, as indicated in Fig. 6. Furthermore, each model was presented with claim events such as Nisynch, Niagree, Alive, Secret, and SKR. Each role communicates with the others using the channels denoted by the words "send" and "recv." If the suggested method is safe, the outcome will prove the veracity of all claims.

VII. Performance Analysis

This section assesses the performance of the proposed protocol and compares it with related protocols such as [15], [16], [17], and [18] in terms of computation time and communication overhead as performance metrics.

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TABLE II
SECUIRTY FEATURES COMPARISON WITH RELEVANT PROTOCOLS

Security property	Hussain et al. [14]	Li et al. [15]	Rabieh et al. [16]	Roman et al. [17]	Hamouid et al. [18]	Gunukula et al. [19]	EV-Auth
Impersonation attack	✓	✓	x	✓	✓	✓	✓
Privileged insider attack	✓	✓	✓	✓	✓	✓	✓
Replay attack	✓	✓	✓	✓	✓	✓	✓
Location privacy	x	✓	x	✓	✓	✓	✓
Man in middle attack	✓	✓	x	✓	✓	✓	✓
DoS attack	✓	x	✓	x	x	x	✓
Forward secrecy	✓	✓	✓	✓	✓	✓	✓
Seamless handover	x	x	x	x	x	x	✓

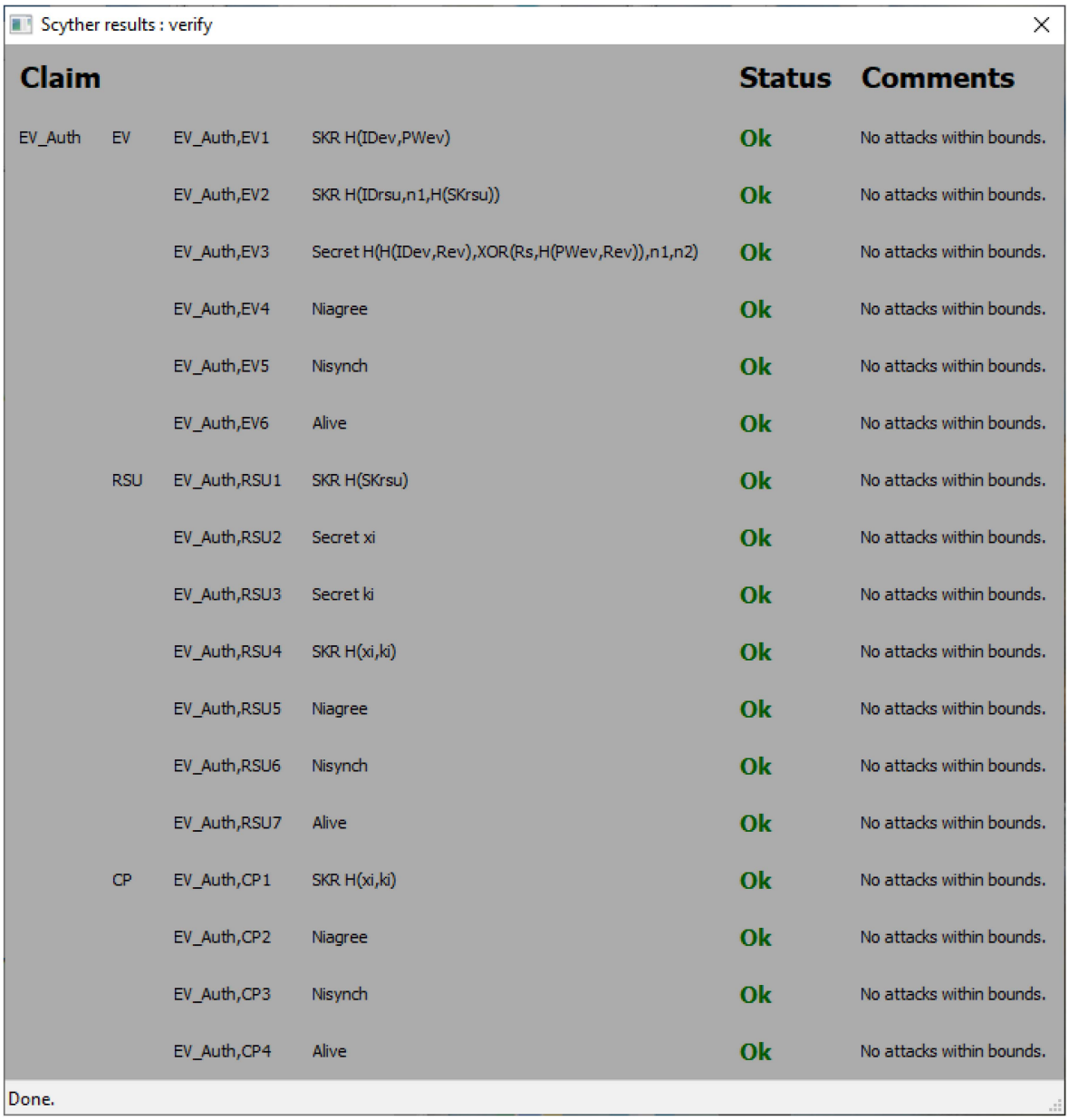


Fig. 6. Verification Results using Scyther.

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TABLE III
COMMUNICATION COST COMPARISON WITH RELEVANT PROTOCOLS

Scheme	Number of messages	Communication overhead(bits)
Rabieh et al. [15]	7	840
Roman et al. [16]	11	752
Hamouid et al. [17]	2	640
Gunukula et al. [18]	9	612
EV-Auth	7	1312

TABLE IV
COMPUTATIONAL TIME COMPARISON WITH RELEVANT PROTOCOLS

Scheme	Required operations	Total time
Rabieh et al. [15]	$8T_{ecm} + 8T_{exp} + 2T_s + 2T_v + 2T_p + 2T_h$	$\approx$ 378.02 ms
Roman et al. [16]	$4T_{ecm} + 4T_p + T_s + T_v + 14T_h$	$\approx$ 242.7 ms
Hamouid et al. [17]	$3T_{ecm} + 2T_e + 3T_h + T_v + T_s$	$\approx$ 54.28 ms
Gunukula et al. [18]	$4T_{exp} + 3T_s + T_v + 15T_h$	$\approx$ 84.18 ms
EV-Auth	$16\mathrm{T}_h$	$\approx$ 5.12 ms

The cost of communication is related to the amount of data sent over the channel. It is determined in bits and is dependent on the total number and kind of parameters exchanged between communicating entities to establish a mutual connection. To calculate the communication cost, the following assumptions about the memory size of each message parameter are used: A 160-bit hash value (SHA-1 [36]), a 32-bit timestamp, a 160-bit identity ID, a 128-bit random nonce, and an elliptic curve point $P = (P_x, P_y)$ of (256 + 256) = 512 bits. The 3072-bit RSA public-key cryptosystem [37] meets the same security requirements as the 256-bit ECC.

Computation time is measured in milliseconds (ms) and is computed based on the total number and kind of cryptographic operations necessary during the communications. If the system requires less computing time, it will be able to send communications between entities more quickly. The computation time is calculated by considering terms and associated time required for T_s – signature generation, T_v – signature verification, T_{ecm} – ECC point multiplication, T_{exp} – modular exponentiation, T_p – bilinear pairing, T_h – one-way hash function, and $T_{(e/d)}$ – encryption/decryption as $T_s \sim 0.6$ ms, $T_v \sim 0.78$ ms [15], $T_{ecm} \sim 17.10$ ms, $T_{exp} \sim 19.20$ ms, $T_p \sim 42.11$ ms [38], $T_h \sim 0.32$ ms [39], 0.32 ms respectively [assumed $T_{(enc/dec} \sim T_h$]. The cost of T_{ER}- exclusive OR is ignored. The Advanced Encryption Standard (AES-128) [40] has been used for symmetric key encryption/decryption. It should be noted that the communication and computational costs vary according to the EV's interaction with several entities present in the dynamic charging station. Additionally, we calculated the communication cost between a single entity in each category, such as an EV, an RSU, and a CP, for precision and ease of understanding.

The computation time and communication costs of protocols proposed by Rabieh *et al.* [15], Roman *et al.* [16], Hamouid *et al.* [17], and Gunukula *et al.* [18] were evaluated. We omitted Hussain *et al.* [13]'s and Li *et al.* [12], [14]'s protocols due to incompatibility between entities and their communications. As illustrated in Tables III and IV, the proposed protocol requires a higher level of communication overhead than comparable protocols, while comparable protocols require more computational power than the proposed protocol.

VIII. CONCLUSION

Several authentication protocols used in EV dynamic charging system are analyzed and discovered that the majority of protocols fall short of meeting security objectives. Then, for an EV dynamic charging system, we proposed a mutual authentication and key agreement protocol. To improve performance, the proposed protocol suite makes use of lightweight cryptographic primitives such as hash functions, Ex-OR operations, and secret keys. Additionally, we included an intriguing feature called seamless handover, which enables EV to more easily roam between multiple RSUs. Additionally, the proposed protocol suite's security has been demonstrated both informally and formally. Thus, the proposed protocol suite can withstand well-known attacks such as man in the middle attacks, replay attacks, forward secrecy attacks, and insider attacks, while also ensuring the location privacy of EV. Additionally, when compared to relevant protocols, the proposed protocol suite demonstrates its efficiency in terms of computation time.

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